

Assignment -2

Subject: Object Oriented Programming in C++

Date of distribution: 2082/03/03

Last Date of submission:2082/03/17

1. How inheritance support reusability features of OOP? Explain with example.
2. How does visibility mode control the access of members in the derived class? Explain with an example.
3. When base class and derived class have the same function name what happens when derived class object calls the function?
4. How does ambiguity occur in multiple inheritance? How can we solve it?
5. What do you mean by ambiguities in inheritance? How can you resolve ambiguities occurs in multipath inheritance?

OR

Explain virtual base class/ diamond problem with suitable examples.

OR

During the time of hybrid inheritance when there is hierarchical inheritance at the upper level and multiple inheritance at the lower level, ambiguity occurs due to duplication of data from the multipath at the grand child class. How this kind of ambiguity can be resolved? Explain with suitable example.

6. Distinguish between subclass and subtype in light of principle of substitutability as well Explain principle of substitutability with suitable examples.
7. How are arguments are sent to base constructors in multiple inheritance? Who is responsible for it.
8. Class 'Y' has been derived from class 'X'. The class 'Y' does not contain any data members of its own. Does the class 'Y' require constructors?
9. What is the major difference between compile time and runtime polymorphism? How compile time and runtime polymorphism can be achieved? Explain with suitable examples.

10. When do you use virtual function? How does it provides runtime polymorphism. Explain with suitable example.

OR

What is virtual function? When and how do we make function virtual explain with suitable example?

11. Differentiate between function overloading and function overriding with suitable examples.

12. Explain importance of polymorphism with suitable examples.

13. How do we make use of a virtual destructor when we need to make sure that the different destructors in inheritance chain are called in order? Explain with an example in C++.

14. Why does this pointer is widely used than object pointer? Explain with examples.

15. Create a base class Shape with a virtual method draw(). Derive two classes Circle and Square, each implementing the draw () method. Demonstrate the principle of substitutability by storing objects of Circle and Square in Shape * pointer. Add an additional method specialFeature() to Circle and explain why it cannot accessed via the Shape* pointer.

16. Write short notes on:

- Deferred method/pure virtual function
- Is a rule/has a rule
- Software reusability
- Virtual inheritance